

IQB Assignment Descriptive Answers

Question 1:

f. Local alignment can ignore mismatching regions at the beginning and end because the **Smith-Waterman** algorithm searches for the best-scoring local region of the two DNA sequences rather than forcing the complete sequence to align. The scoring matrix is initialised with zeros, and during matrix filling, 0 is also included in the max function to get possible score. Hence, if alignment on continuation reduces the score, the algorithm can reset this score to zero and start a new local alignment.

During traceback, the process stops when a cell with zero is reached. Hence, sequence regions before the starting point or after the ending point of the optimal local alignment can remain unmatched without affecting the alignment score.

Question 2:

d. The global alignment must consider the entire length of both DNA sequences. The **Needleman-Wunsch** algorithm starts the traceback from the bottom-right cell of the scoring matrix and continues until both sequences reach their beginning. Any bases that do not have a corresponding base at the beginning or end of the other sequence can not be ignored. They must be added to the alignment using gaps, and the corresponding gap penalties should be also included in the final alignment score.

Similarly, mismatches at the ends are also considered and contribute their mismatch penalty. Thus, global alignment provides an alignment that covers the complete lengths of both sequences including any necessary gaps and mismatches at the ends.

